DS	**Session**	Searching	**Question Information**	Level 1 Challenge 5

Problem Description:
Nesamani wants to paint a beautiful picture on a building that is M sections long. Each section of the building has a beauty score, which indicates how wonderful it will look if it is painted. Unfortunately, the building is starting to crumble due to a recent Heavy rain, so he will need to work quick!

At the beginning of each day, Nesamani will paint one of the sections of the building. On the first day, he is free to paint any section he likes. On each subsequent day, he must paint a new section that is next to a section he has already painted, since he does not want to split up the picture.

At the end of each day, one section of the building will be destroyed. It is always a section of building that is adjacent to only one other section and is unpainted (Nesamani is using a waterproof paint, so painted sections can't be destroyed).

The total beauty of Nesamani's picture will be equal to the sum of the beauty scores of the sections he has painted. Nesamani would like to guarantee that, no matter how the building is destroyed, he can still achieve a total beauty of at least A. What's the maximum value of A for which he can make this guarantee?

Constraints:
$1 \leq T \leq 100$.
$2 \leq M \leq 100$.

Input Format:
The first line of the input gives the number of test cases, T. Each test case starts with a line containing an integer M. Then, another line follows containing a string of M digits from 0 to 9. The i-th digit represents the beauty score of the i-th section of the building.

Output Format:
Print the output in a separate lines contains the maximum beauty score that Nesamani can guarantee that he can achieve, as described above.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4
4
1332
4
9583
3
616
10
1029384756
```

EXPECTED OUTPUT

```
6
14
7
31
```

Test Case 2

INPUT (STDIN)

```
4
3
332
3
583
4
1616
8
29384756
```

EXPECTED OUTPUT

```
6
13
7
24
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
for(k=1;k<=T;++k)
```

Test Case 2

KEYWORD

```
vector<int> b(N+1);
```

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
4	175	25

Code

solution.c

```c
#include <stdio.h>

int main() {
    int T;
    if (scanf("%d", &T) != 1) return 0;
    for (int k = 1; k <= T; ++k) {
        int N;
        char s[105];
        scanf("%d %s", &N, s);
        int b[105];
        for (int i = 0; i < N; i++) b[i] = s[i] - '0';
        int kLen = (N + 1) / 2;
        int maxScore = 0;
        int curSum = 0;
        for (int i = 0; i < kLen; i++) curSum += b[i];
        maxScore = curSum;
        for (int i = kLen; i < N; i++) {
            curSum += b[i] - b[i - kLen];
            if (curSum > maxScore) maxScore = curSum;
        }
        printf("%d\n", maxScore);
    }
    return 0;
}
```

Output

